

Gosford High School

Year 11

2007

Preliminary Higher School Certificate

Mathematics

Assessment Task 2

Time Allowed - 60 minutes
(plus 5 minutes reading time)

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

QUESTION 1 (12 marks)

- (i) 35% of Jamiel's weekly wage is spent on rent. Find his weekly wage if his rent is \$128.80. (1)
- (ii) Write $0.\dot{7}1\dot{5}$ as a fraction in simplest form (2)
- (iii) Find the value of a given that $\sqrt{45} + \sqrt{80} = a\sqrt{5}$ (2)
- (iv) Simplify $\frac{24 + \sqrt{72}}{6}$ (1)
- (v) $\frac{2}{7}, \sqrt{7}, -10, 0.6, 4\frac{1}{2}\%, \sqrt{-4}, \pi, |16|$
From the above set of numerals list the set of Rational numbers (1)
- (vi) Simplify $\sqrt{(-6)^2} - |-6|$ (1)
- (vii) Find the value of $\frac{\sqrt{21 \cdot 68 - 14 \cdot 9}}{2 \cdot 4\pi}$ correct to 2 significant figures (1)
- (viii) Simplify $x - \frac{1}{x}$ if $x = \frac{1}{\sqrt{2} - 1}$ (3)

QUESTION 2 (12 marks)

- (i) Factorise (a) $2a^3 + 16$ (2)
- (b) $1 - 2a - 24a^2$ (1)
- (ii) Solve (a) $8p^3 + 64 = 0$ (1)
- (b) $\frac{3}{2a} + \frac{1}{a} = 4 - \frac{5}{3a}$ (2)
- (c) $|2x + 1| < 7$ (2)
- (iii) Find the exact roots of $2m^2 + 3m - 1 = 0$ (2)
- (iv) Make x the subject of the equation $4x - 3xy - 12 = 0$ (2)

QUESTION 3

(12 marks)

- (i) Simplify $\frac{3}{x^2 - x - 2} - \frac{4}{x^2 - 2x - 3}$ (3)
- (ii) Simplify $\frac{xm^{-1} + mx^{-1}}{x^2 + m^2}$ (3)
- (iii) Simplify $\frac{8^{2n} \times 27^n}{18 \times 12^{3n}}$ (3)
- (iv) Solve equation $2x - 3 = \sqrt{33 - 2x}$ (3)

QUESTION 4

(12 marks)

- (i) Plot the points A(-4, 5), B(4, -1) and C(^S~~0~~, 3) on a number plane. Label the axes and the points clearly and neatly on your diagram. (2)
- (ii) Find the midpoint of the interval CA. (1)
- (iii) Find the length of the interval AB. (1)
- (iv) Find the gradient of AB (1)
- (v) Find the equation of the line AB, writing your answer in general form. (2)
- (vi) On your diagram draw the perpendicular from C to the line AB (1)
- (vii) Find the perpendicular distance from the point C to the line AB. (2)
- (viii) Find the area of the triangle ABC. (2)

QUESTION 5

(12 marks)

The line RT has equation $4x + 3y - 6 = 0$ and the line QT has equation $2x - 4y + 3 = 0$.

- (i) Find the gradient of the line RT. (1)
- (ii) Find the angle of inclination of the line RT (answer to the nearest degree). (2)
- (iii) Find the equation of the line that is perpendicular to RT and passing through the point $(-3, 1)$ (2)
- (iv) The line QT cuts the x and y axes at M and N respectively. Find the coordinates of M and N. (2)
- (v) The point $(-17, k)$ lies on the line QT. Find the value of k . (2)
- (vi) Find the coordinates of the point T. (3)

QUESTION 6

(12 marks)

- (i) Graph on a number plane the line $y = 2 - \frac{x}{4}$. (2)
- (ii) On your diagram shade the region that simultaneously satisfies the inequalities
$$y \leq 2 - \frac{x}{4},$$
$$x + 2 > 0 \quad \text{and}$$
$$y \geq 0$$
(4)
- (iii) On a new diagram sketch the region satisfying $0 \leq x + y < 4$ (3)
- (iv) The line $3x - 4y + f = 0$ is 2 units from the point A $(1, -3)$. Find the value(s) of f . (3)

QUESTION 1

2007

$$(i) 0.35 \times \text{Wage} = \$128.80$$

$$\text{Wage} = \frac{\$128.80}{0.35}$$

$$= \$368$$

$$(ii) \text{ let } x = 0.715$$

$$\therefore 100x = 71.515$$

$$99x = 70.8$$

$$x = \frac{70.8}{99}$$

$$x = \frac{708}{990}$$

$$x = \frac{118}{165}$$

$$(iii) \sqrt{45} + \sqrt{80} = a\sqrt{5}$$

$$3\sqrt{5} + 4\sqrt{5} = a\sqrt{5}$$

$$7\sqrt{5} = a\sqrt{5}$$

$$\therefore a = 7$$

$$(iv) \frac{24 + \sqrt{72}}{6} = \frac{24 + 6\sqrt{2}}{6}$$

$$= \frac{6(4 + \sqrt{2})}{6}$$

$$= 4 + \sqrt{2}$$

$$v) \frac{2}{7}, -10, 0.6, 4\frac{1}{2}\%, |16|$$

$$vi) 6 - 6 = 0$$

$$vii) 0.35$$

$$viii) x = \frac{1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \sqrt{2}+1$$

$$x - \frac{1}{x} = \sqrt{2}+1 - (\sqrt{2}-1)$$

$$= \sqrt{2}+1 - \sqrt{2}+1$$

$$= 2$$

QUESTION 2

$$i) a) 20^3 + 16 = 2(a^3 + 8)$$

$$= 2(a+2)(a^2 - 2a + 4)$$

$$b) 1 - 2a - 24a^2$$

$$= (1 - 6a)(1 + 4a)$$

OR.

$$= -(6a - 1)(4a + 1)$$

$$ii) a) 8p^3 = -64$$

$$p^3 = -8$$

$$p = -2$$

$$b) \frac{3}{2a} + \frac{1}{a} = 4 - \frac{5}{3a}$$

$$9 + 6 = 24a - 10$$

$$25 = 24a$$

$$\frac{25}{24} = a$$

$$c) |2x+1| < 7$$

$$-7 < 2x+1 < 7$$

$$-8 < 2x < 6$$

$$-4 < x < 3$$

$$(iii) 2m^2 + 3m - 1 = 0$$

$$m = \frac{-3 \pm \sqrt{9 - 4(2)(-1)}}{4}$$

$$= \frac{-3 \pm \sqrt{17}}{4}$$

$$(iv) 4x - 3xy = 12$$

$$x(4 - 3y) = 12$$

$$x = \frac{12}{4 - 3y}$$

QUESTION 3

$$i) \frac{3}{(x-2)(x+1)} - \frac{4}{(x-3)(x+1)}$$

$$= \frac{3(x-3) - 4(x-2)}{(x-2)(x+1)(x-3)}$$

$$= \frac{3x - 9 - 4x + 8}{(x-2)(x+1)(x-3)}$$

$$= \frac{-x - 1}{(x-2)(x+1)(x-3)}$$

$$= \frac{-(x+1)}{(x-2)(x+1)(x-3)}$$

$$= \frac{-1}{(x-2)(x+1)(x-3)}$$

$$ii) \frac{x}{m} + \frac{m}{x}$$

$$= \frac{x^2 + m^2}{xm}$$

Multiply top & bottom by xm

$$= \frac{x^2 + m^2}{xm(x^2 + m^2)}$$

$$= \frac{1}{xm}$$

Part (iii) at end

$$iv) (2x-3)^2 = 33 - 2x$$

$$4x^2 - 12x + 9 = 33 - 2x$$

$$4x^2 - 10x - 24 = 0$$

$$2x^2 - 5x - 12 = 0$$

$$(2x+3)(x-4) = 0$$

$$x = -\frac{3}{2}, 4$$

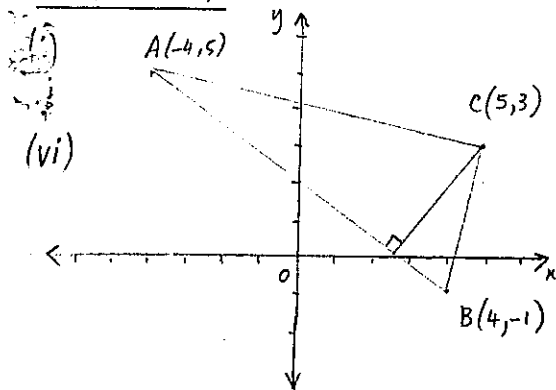
But when $x = -\frac{3}{2}$

$$L.H.S. = -6, R.H.S. = 6$$

$\therefore x = -\frac{3}{2}$ is not a solution

ie $x = 4$ is only solution

QUESTION 4



(ii) Midpoint = $\left(\frac{5+(-4)}{2}, \frac{3+5}{2}\right)$
 $= \left(\frac{1}{2}, 4\right)$

(iii) $AB = \sqrt{(4-(-4))^2 + (-1-5)^2}$
 $= \sqrt{8^2 + (-6)^2}$
 $= \sqrt{64+36}$
 $= \sqrt{100}$
 $= 10$

(iv) $m_{AB} = \frac{-1-5}{4-(-4)}$
 $= \frac{-6}{8}$
 $= -\frac{3}{4}$

(v) Equation of AB is
 $y+1 = -\frac{3}{4}(x-4)$
 $4y+4 = -3x+12$
 $3x+4y-8=0$

vii) $d = \frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$
 $= \frac{|3(5)+4(3)-8|}{\sqrt{3^2+4^2}}$
 $= \frac{|19|}{5}$
 $= \frac{19}{5} \text{ or } 3\frac{4}{5}$

viii) Area of $\triangle ABC = \frac{1}{2}bh$
 $= \frac{1}{2} \times 10 \times \frac{19}{5}$
 $= 19 \text{ sq units.}$

QUESTION 5

i) $3y = -4x+6$
 $y = -\frac{4x}{3}+2$

Gradient of RT is $-\frac{4}{3}$

ii) Let θ be angle of inclination
 $\therefore \tan \theta = -\frac{4}{3}$
 $\theta = 127^\circ$

iii) Equation of perpendicular
is $y-1 = \frac{3}{4}(x+3)$
 $4y-4 = 3x+9$
 $3x-4y+13=0$

(iv) at M, $y=0$
 $\therefore 2x = -3$
 $x = -\frac{3}{2} \rightarrow M\left(-\frac{3}{2}, 0\right)$
at N, $x=0$
 $\therefore -4y = -3$
 $y = \frac{3}{4} \rightarrow N\left(0, \frac{3}{4}\right)$

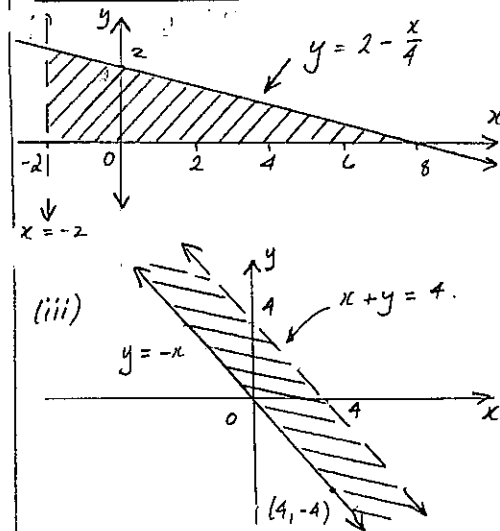
v) $(-17, k)$ satisfies
 $\therefore 2(-17) - 4k + 3 = 0$
 $-34 - 4k + 3 = 0$
 $-4k = 31$
 $k = -\frac{31}{4} \text{ or } -7\frac{3}{4}$

vi) $4x+3y-6=0 \dots (RT)$
 $4x-8y+6=0 \dots (QT)$
 $\therefore 11y-12=0$
 $11y=12$
 $y = \frac{12}{11}$

$\therefore 2x - 4\left(\frac{12}{11}\right) + 3 = 0$
 $22x - 48 + 33 = 0$
 $22x = 15$
 $x = \frac{15}{22}$

T is the point $\left(\frac{15}{22}, \frac{12}{11}\right)$

QUESTION 6



(iii)
(iv) Perp. distance = 2
 $\therefore \frac{|3(1)+(-4)(-3)+f|}{\sqrt{3^2+4^2}} = 2$
 $\therefore \frac{|15+f|}{5} = 2$
 $|15+f| = 10$
 $15+f = \pm 10$
 $f = -5, -25$

Question 3 (iii) (Sorry)
 $\frac{2n}{8 \times 27} = \frac{2^n}{18 \times (2^2 \times 3)^{3n}}$
 $= \frac{2^n}{18 \times 2^{6n} \times 3^{3n}}$
 $= \frac{1}{18}$